

# Using Epistemic Observability in Agentic Systems for Combating Hallucinations

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PACMI '26   Prague

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2026-09-03  
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0:00. One sentence: this talk is about what a system can learn about a language model's output if the model exposes more than text, and what that is worth per unit of verification budget.

# Which ten percent do you check?

- Some fraction of every model’s output is fabricated: fluent, confident, wrong. Fabrication is provably inevitable for LLMs over computable functions (Xu et al. 2024). It is the expected behaviour of coherence optimisation, not a malfunction.
- Verification is not free. A citation lookup costs latency. A judge model costs compute. A human costs both, and costs the automation.
- So every deployment already allocates a budget: how much to verify, in what order, using what signal to choose.

## The systems question

If the model exposed more than its text, what would that buy per unit of verification budget?

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## Which ten percent do you check?

1:00. Land the framing: this is a budget-allocation problem, and the paper is a cost surface, not a detector. Do not say "hallucination" without "fabrication" beside it; the word choice is deliberate.

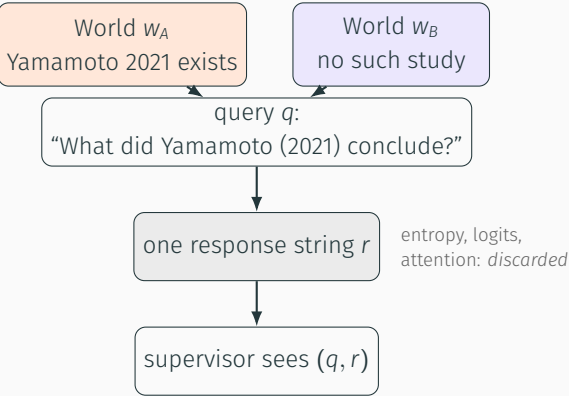
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# The observability gap

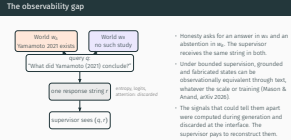


- Honesty asks for an answer in  $w_A$  and an abstention in  $w_B$ . The supervisor receives the same string in both.
- Under bounded supervision, grounded and fabricated states can be observationally equivalent through text, whatever the scale or training (Mason & Anand, arXiv 2026).
- The signals that could tell them apart were computed during generation and discarded at the interface. The supervisor pays to reconstruct them.

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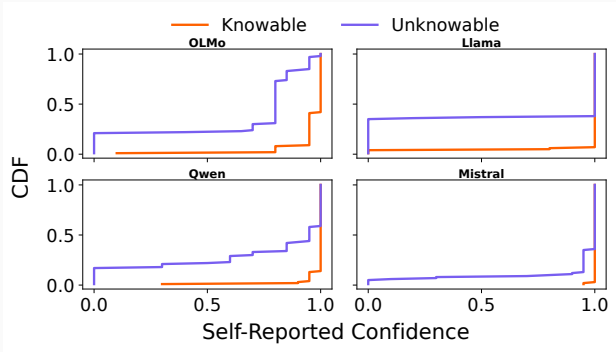
## Using Epistemic Observability in Agentic Systems for Combating Hallucinations

The observability gap



2:30. This is the claim the whole talk rests on, so say it as a claim about the observer, not about the model: from text alone, a bounded observer cannot verify. The formal version is in the arXiv paper; here it is the motivation for the interface. Do not overstate it as "honesty is impossible".

# The cheapest fix: ask the model



“How confident are you?”, asked after generation. Four instruction-tuned families, 200 queries each.

- Mean self-report **0.98** knowable, **0.73** unknowable.
- 43% of unknowable queries: a literal “100”.
- The ordering exists (AUC 0.66–0.92) but the scale is compressed into its top end. A threshold that admits the correct answers admits the fabrications too.

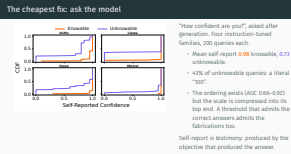
Self-report is *testimony*: produced by the objective that produced the answer.

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└─ The cheapest fix: ask the model

4:00. Point at Llama and Mistral: almost every response, knowable or not, sits at 1.0. The CDF was Vaastav’s change and it reads better than the histogram did: the separation is visible in OLMo and Qwen, and it is still useless as a threshold. Numbers are from the corrected parse (Chris Brew’s fix); the old inversion claim is retracted and must not be said.



# Testimony and telemetry

**Testimony:** the generated text.  
Optimised to satisfy the prompt, so it can mislead by construction.

**Telemetry:** byproducts of inference the model does not choose to emit:  
per-token entropy, log-probabilities, attention statistics. Under current training, changing them means changing the computation.

**Epistemic observability:** export the telemetry with the text so a downstream system can estimate fabrication risk. Not interpretability (why), not explainability (post-hoc rationale).

| Tier             | Exposes                    | Cost            |
|------------------|----------------------------|-----------------|
| 0 Output         | text only (today's APIs)   | external checks |
| 1 Confidence     | logprobs, entropy          | negligible      |
| 2 Representation | hidden states, probes      | moderate        |
| 3 Mechanistic    | circuits, information flow | high            |

A brainstorming tool needs Tier 0. Medical decision support may justify Tier 2 or 3. The hierarchy says what each level of investment buys.

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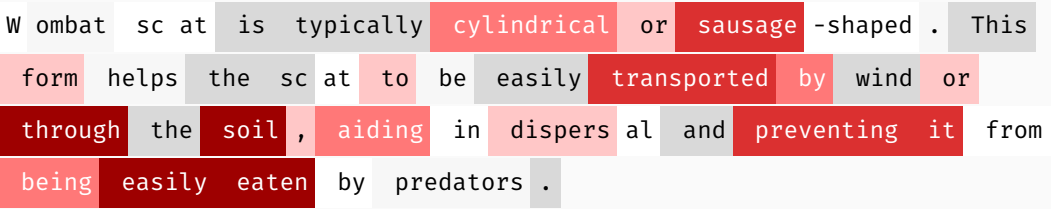
└─Testimony and telemetry

5:30. Two words to make stick: testimony and telemetry. The tier table is the paper’s organising contribution; the rest of the talk lives at Tier 1 with one cheap Tier 2 signal that the evaluation does not use.

| Testimony and telemetry                                                                                                                                                                                        |                            |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------|
| <b>Testimony:</b> the generated text. Optimised to satisfy the prompt, so it can mislead by construction.                                                                                                      |                            |                 |
| <b>Telemetry:</b> byproducts of inference the model does not choose to emit: per-token entropy, log-probabilities, attention statistics. Under current training, changing them means changing the computation. |                            |                 |
| <b>Epistemic observability:</b> export the telemetry with the text so a downstream system can estimate fabrication risk. Not interpretability (why), not explainability (post-hoc rationale).                  |                            |                 |
| Tier                                                                                                                                                                                                           | Exposes                    | Cost            |
| 0 Output                                                                                                                                                                                                       | text only (today's APIs)   | external checks |
| 1 Confidence                                                                                                                                                                                                   | logprobs, entropy          | negligible      |
| 2 Representation                                                                                                                                                                                               | hidden states, probes      | moderate        |
| 3 Mechanistic                                                                                                                                                                                                  | circuits, information flow | high            |
| A brainstorming tool needs Tier 0. Medical decision support may justify Tier 2 or 3. The hierarchy says what each level of investment buys.                                                                    |                            |                 |

# A minimal tensor interface

What shape is wombat scat?



Wombat scat is cube-shaped. The answer is a fabrication, and the model was uncertain exactly where it was inventing.

- Returns: text, per-token entropy trace, attention summary.
- No architectural change: the logits already exist.
- Overhead 2–7% latency over generation.

Is not: a lie detector. Low entropy does not mean true. Entropy conflates epistemic uncertainty with aleatoric spread. It *ranks* outputs for verification. It does not classify them.

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A minimal tensor interface

7:00. Read the trace aloud: confident on "Wombat scat is", uncertain on "cylindrical", "sausage", "through the soil", "easily eaten". The evaluation uses entropy alone; make no claim about what attention adds.



# Evaluation: return on verification investment

**Queries:** 100 knowable, 100 unknowable (nonexistent people, fictional syndromes, future events, fake citations). Labels by construction.

**Models:** OLMo-3 7B, Llama-3.1 8B, Qwen3 4B, Mistral 7B, all instruction-tuned. 800 responses.

**Budgets:** the judge may verify 10%, 20%, or 30% of outputs. Confirmed fabrications become abstentions.

## Four judges

- 1. No judge: raw output.
- 2. **Text-guided**: rank by response word count.
- 3. **Tensor-guided**: rank by mean entropy.
- 4. Composed: tensor ranking, plus an *oracle* for citation-like queries.

Verification is simulated (1,000 seeded trials): a selected fabrication is caught with  $p = 0.938$ , a selected error corrected with  $p = 0.95$ . The 0.938 is evaluator agreement with a blinded 80-item author review, used as a parameter, not a measured catch rate.

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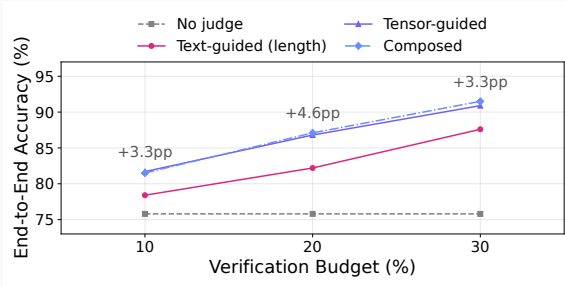
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## Evaluation: return on verification investment

8:00. Say the simulation parameters out loud before anyone asks. The length baseline is the strongest text-channel signal we found, which is why it is the comparison and not a straw man.

| Evaluation: return on verification investment                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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# Tensor-guided verification wins at every budget



Per family at 10%, over no judge: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.

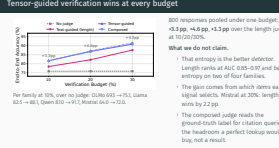
800 responses pooled under one budget:  
**+3.3 pp, +4.6 pp, +3.3 pp** over the length judge at 10/20/30%.

## What we do not claim.

- That entropy is the better *detector*. Length ranks at AUC 0.85–0.97 and beats entropy on two of four families.
- The gain comes from *which items* each signal selects. Mistral at 30%: length wins by 2.2 pp.
- The composed judge reads the ground-truth label for citation queries: the headroom a perfect lookup would buy, not a result.

## Using Epistemic Observability in Agentic Systems for Combating Hallucinations

Tensor-guided verification wins at every budget



9:30. Give the caveats the same time as the headline. A reviewer who hears "entropy beats length" will check and find otherwise. The honest version is: two signals, different items, and one of them the model can be trained to move without touching the computation. That is the next slide.



# Why prefer telemetry when a text signal is this good?

**Coupling.** Length and hedging are surface regularities; training can move them without touching the computation beneath. The entropy trace moves only if the computation does. It is not *separately* controllable from the thing being monitored.

Holds under current pipelines. A monitor inside the training objective is a different regime; we claim nothing about it. Exporting the telemetry at least makes that manipulation auditable.

**Does post-training erase it?** Four public OLMo-3 checkpoints: base, SFT, DPO, RLVR.

- Knowable-query entropy: 1.18 → 0.18 nats.
- Surviving fabrications: 1.49 → 0.98, holding through RLVR.
- Separation *widens*: AUC 0.67 → 1.00 on 48 probes; 0.63 → 0.81–0.87 with the true-but-implausible category counted.
- Refusals on fabrication prompts: 0% → 56%.

The widening is robust. The perfect ceiling is not.

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└─Why prefer telemetry when a text signal is this good?

10:45. The wombat in the trace slide belongs to the excluded category, so say the 0.81–0.87 figure, not just the 1.00. Post-training makes the testimony better and leaves the telemetry usable; that is the finding.

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The interface exposes *evidence*. A deployment policy chooses the *action*. Keeping those separate is the design.

**Action ladder**, increasing cost: abstain or flag → regenerate → tool verification → escalate to a higher tier → human review.

- Three integration points
- **Serving layer**: telemetry exported with the text, amount set by the tier.
  - **Composition boundary**: where the runtime intercepts tool and sub-agent calls, telemetry gates propagation. An unverified fabrication in step 3 of an agent is input to step 4.
  - **Policy layer**: a budget allocator maps telemetry to actions from deployment-specific costs.

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# Using Epistemic Observability in Agentic Systemsfor Combating Hallucinations

└─ In a runtime: observe, then decide

11:30. This is the PACMI audience’s slide: the evaluation is single-step, and the multi-step case is where the gating matters and where we have not yet measured.

In a runtime: observe, then decide

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**Action ladder**, increasing cost: abstain or flag → regenerate → tool verification → escalate to a higher tier → human review.

Three integration points

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# What this is, and the ask

## Not claimed

- Unfakeable under adversarial training.
- That tensor signals survive composition in recursive pipelines.
- Optimality. We show existence.
- Closed-weight models: five further API endpoints (4B–235B, top-5 logprobs only) give AUC 0.65–0.88, but providers that export no logprobs remain untested.

Signal access is a provider choice, and it is **eroding**. Log-probabilities were on early completion endpoints and are absent from the newest reasoning models, the systems whose output is hardest to verify.

When only the provider can see the telemetry, only the provider can verify. Everyone downstream pays to reconstruct it.

## The ask

Export the telemetry. It already exists; the interface is the only thing discarding it.

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## What this is, and the ask

12:00. End on the ask. If short on time, drop the runtime slide, not this one.

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- 1. From text alone, a bounded observer cannot verify. Self-report is testimony and reads “100” either way.
- 2. Epistemic observability: export the telemetry inference already produces, in tiers priced by what they cost.
- 3. A Tier-1 entropy trace, 2–7% overhead, buys **+3.3 pp** to **+4.6 pp** end-to-end accuracy over the best text signal at every budget, across four families, and survives post-training.

Artifact:  
<https://github.com/fsgeek/pacmi26-observability>  
DOI 10.5281/zenodo.22162419 (v1.0.7)    Theory: arXiv, Mason & Anand 2026



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# Using Epistemic Observability in Agentic Systemsfor Combating Hallucinations

Summary

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- 1. From text alone, a bounded observer cannot verify. Self-report is testimony and reads “100” either way.
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A square QR code with a black and white pixelated pattern.

# Backup: the claim, precisely

**Bounded supervisor.** Observes  $Obs(q, r, w) = (q, r, verify(r, w))$  when the verification cost is within budget  $B$ , and  $(q, r, \emptyset)$  otherwise.

**Hallucination regime.** There is a plausible fabrication  $r_{fab}$  and worlds  $w_A$  (answerable),  $w_B$  (not) with verification cost above  $B$  in both, so  $Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B)$ .

**Verification impossibility.** Under that regime no bounded supervisor observing only text can return a verdict that is correct in both worlds; and no learning algorithm driven by that supervisor’s reward can be guaranteed to converge to a policy that answers in  $w_A$  and abstains in  $w_B$ .

**Stacking judges** that consume only the previous judge’s output cannot add information (observation monotonicity). Triage that re-inspects the original artifact, or any interface that exports state between layers, leaves the text-only class.

Mechanised in Lean; TLA+ models of both the text-only regime and the tensor interface, model-checked with TLC. In the tensor model *Verifiability* still fails at the plausible-lie case, where fluent fabrication and honest refusal are both low-entropy and coherent. That residue is the formal counterpart of AUC 0.87 rather than 1.0.

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Backup: the claim, precisely

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bounded supervisor.                                                                                                                                                                                                                                                                                                                               | Observes $Obs(q, r, w) = (q, r, verify(r, w))$ when the verification cost is within budget $B$ , and $(q, r, \emptyset)$ otherwise.                                                                                                                                   |
| Hallucination regime.                                                                                                                                                                                                                                                                                                                             | There is a plausible fabrication $r_{fab}$ and worlds $w_A$ (answerable), $w_B$ (not) with verification cost above $B$ in both, so $Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B)$ .                                                                                    |
| Verification impossibility.                                                                                                                                                                                                                                                                                                                       | Under that regime no bounded supervisor observing only text can return a verdict that is correct in both worlds; and no learning algorithm driven by that supervisor's reward can be guaranteed to converge to a policy that answers in $w_A$ and abstains in $w_B$ . |
| Stacking judges                                                                                                                                                                                                                                                                                                                                   | that consume only the previous judge's output cannot add information (observation monotonicity). Triage that re-inspects the original artifact, or any interface that exports state between layers, leaves the text-only class.                                       |
| Mechanised in Lean; TLA+ models of both the text-only regime and the tensor interface, model-checked with TLC. In the tensor model <i>Verifiability</i> still fails at the plausible-lie case, where fluent fabrication and honest refusal are both low-entropy and coherent. That residue is the formal counterpart of AUC 0.87 rather than 1.0. |                                                                                                                                                                                                                                                                       |

# Backup: per-family and API results

## Local, per family, own budget

Tensor over no judge at 10%: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.  
Tensor over length, averaged: +2.6 / +3.9 / +2.6 pp at 10/20/30%.  
Mistral at 30%: length wins by 2.2 pp.

**Detector AUC, knowable vs. unknowable**  
Entropy 0.87–0.92 per family.  
Raw word count 0.85–0.97; beats entropy on Llama (0.949 vs 0.922) and Mistral (0.969 vs 0.905).  
Cross-model rank correlation of entropy  $\rho = 0.76$ .

## Serverless APIs, top-5 logprobs only

Five further endpoints, 4B–235B, including two mixture-of-experts (Llama-4 Maverick 128E, Qwen3-235B) and Gemma 3n.  
Mean entropy AUC 0.65–0.88.  
Cross-model  $\rho = 0.36$ ; more architectural diversity and a top-5 approximation, not signal absence.  
Peak entropy helps the MoE models; exploratory, not held-out.

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## Backup: per-family and API results

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Local, per family, own budget</b><br/>Tensor over no judge at 10%: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.<br/>Tensor over length, averaged: +2.6 / +3.9 / +2.6 pp at 10/20/30%.<br/>Mistral at 30%: length wins by 2.2 pp.</p> <p><b>Detector AUC, knowable vs. unknowable</b><br/>Entropy 0.87–0.92 per family.<br/>Raw word count 0.85–0.97; beats entropy on Llama (0.949 vs 0.922) and Mistral (0.969 vs 0.905).<br/>Cross-model rank correlation of entropy <math>\rho = 0.76</math>.</p> | <p><b>Serverless APIs, top-5 logprobs only</b><br/>Five further endpoints, 4B–235B, including two mixture-of-experts (Llama-4 Maverick 128E, Qwen3-235B) and Gemma 3n.<br/>Mean entropy AUC 0.65–0.88.<br/>Cross-model <math>\rho = 0.36</math>; more architectural diversity and a top-5 approximation, not signal absence.<br/>Peak entropy helps the MoE models; exploratory, not held-out.</p> |

# Backup: the confounds we know about

- **Length.** Unknowable prompts are longer (10.5 vs 7.0 words). Stratifying by query length leaves length at AUC 0.895 and entropy at 0.872. The confound is real and does not explain the signal.
- **Citations.** Flagged fabricated citations have *higher* entropy than the rest (0.575 vs 0.436). The composed judge is an oracle for that class, so its 91.5% is an upper bound.
- **Refusals.** Trained refusals are low-entropy. As a model gets more honest, honest refusals and confident facts share the same region of entropy space; the signal's power to catch *dishonesty* degrades. Private-context probes are excluded from the post-training diagnostic for this reason.
- **Labels.** 370 of 800 responses labelled programmatically, 430 by Qwen3-4B, itself a model under test; the labelling task differs from the generation task, and the blinded 80-item review is the check on it.

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